Bit-Width Scaling Laws for Mathematical Reasoning in Small Language Models

Luis Benvenuto[1]

[1]Data Science & Mathematics, Independent Research

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Abstract

We study how post-training quantization of the weights of a small autoregressive language model affects its ability to approximate elementary mathematical functions. A dependency-free per-channel symmetric quantizer is applied at bit-widths $\{32, 16, 8, 4, 2\}$, where 2-bit corresponds to a ternary BitNet-style approximation. On the open-weight Qwen2.5-0.5B model evaluated on a synthetic arithmetic benchmark, we observe the expected monotonic degradation: logit MSE versus the FP32 reference spans many orders of magnitude across the sweep. Layer-wise submodule sensitivity reveals that attention and feed-forward weights contribute asymmetrically to the residual error. The pipeline is configuration-driven and rescales to GSM8K, the Hendrycks MATH benchmark, and 7B-parameter mathematical models behind a GPU configuration.

1 Introduction

Quantization is now the dominant technique for deploying large language models on commodity hardware. Yet whilst memory and latency benefits are well-understood, the question of *which capabilities degrade first* under aggressive precision reduction remains open. Mathematical reasoning is a natural stress test: arithmetic and symbolic problems require many sequential non-linear approximations and amplify errors quickly.

Question. How does the residual error in non-linear function approximation by a small open-weight language model scale with the bit-width of its parameters, and which submodules—attention or feed-forward—contribute most to that error?

Contributions. (i) A dependency-free per-channel symmetric quantizer covering FP32, FP16, INT8, INT4, and ternary (1.58-bit) regimes. (ii) A reproducible evaluation pipeline combining a synthetic arithmetic benchmark with optional GSM8K loading. (iii) A submodule-sensitivity test isolating attention from feed-forward contributions to logit error. (iv) An end-to-end paper whose numbers and figures are regenerated from raw results on each run; the same code rescales to TinyLlama-1.1B and Qwen2.5-Math-7B via the GPU configuration.

2 Related Work

Post-training quantization (PTQ). GPTQ [2] and AWQ [4] achieve nearly loss-less INT4 weights via second-order error correction and activation-aware scaling. SmoothQuant [6] migrates activation outliers into the weights. We deliberately use a simpler per-channel symmetric quantizer to establish a clear lower bound on the quality–bit-width trade-off; better quantizers are a strict improvement, not a contradiction.

Ternary and BitNet. BitNet [5] demonstrates that training from scratch with ternary weights $\{-1, 0, +1\}$ can match FP16 baselines at sufficient scale. Our 2-bit setting is post-training and is best interpreted as a stress test, not a recipe.

Mathematical capabilities of LLMs. GSM8K [1] and the Hendrycks MATH benchmark [3] are the canonical evaluations. Our local configuration uses synthetic arithmetic problems so that the pipeline runs deterministically without network access; the GPU configuration substitutes GSM8K and MATH subsets.

3 Methodology

3.1 Quantizer

Let $W \in \mathbb{R}^{n_{\text{out}} \times n_{\text{in}}}$ be the weight of a linear layer. We use the following per-row symmetric quantizer:

- **32-bit:** identity.

- **16-bit:** cast to FP16, round-trip back to working dtype.
- **8 / 4-bit:** symmetric per-row scale $s_i = \max_j |W_{ij}|/q_{\max}$, with $q_{\max} \in \{127, 7\}$ and the quantized weight $W'_{ij} = s_i \cdot \mathrm{clip}\big(\mathrm{round}(W_{ij}/s_i), \pm q_{\max}\big)$.
- **2-bit (ternary):** per-row threshold $\tau_i = 0.7\,|\bar{W}|_i$, where $|\bar{W}|_i$ is the row's mean absolute value. Weights below the threshold are zeroed; surviving weights take values $\pm s_i$, where s_i is the mean magnitude of surviving weights. This is a deterministic, dependency-free approximation of BitNet's ternary regime.

3.2 Logit divergence

For each problem we record the model's last-token logits under FP32 and under each quantization regime, and compute MSE $= \mathbb{E}\big[(\ell^{\mathrm{ref}} - \ell^{q})^2\big]$ and the forward KL divergence $\mathrm{KL}(p \,\|\, q)$ between the corresponding softmax distributions. MSE measures raw drift; KL is sensitive to the decisive directions of the distribution.

3.3 Submodule sensitivity

We isolate attention vs. feed-forward error contributions by applying quantization *only* to layers whose module path contains {`attn`, `attention`, `self_attn`} or {`mlp`, `ffn`, `feed_forward`, `fc`} respectively, holding the complement at FP32. The bit-width used for this experiment is 4, the regime where the global sweep shows the steepest accuracy degradation.

4 Experiments

Model. Qwen2.5-0.5B (the open-weight 494M-parameter Qwen model). The GPU configuration switches to TinyLlama-1.1B-Chat-v1.0 and Qwen2.5-Math-7B without code changes.

Benchmark. A deterministic synthetic arithmetic benchmark of 80 items spanning addition, subtraction and multiplication with operands sampled uniformly from $\{1, \ldots, 100\}$, plus optional GSM8K subset behind `gpu_full.yaml`. Accuracy is judged by string-matching the first integer in the model's greedy completion.

Procedure. For each bit-width $b \in \{32, 16, 8, 4, 2\}$ we deep-copy the FP32 model, apply the quantizer in place, generate answers, and compute logit MSE and KL on a held-out set of 50 items. Each step runs in a few minutes on CPU/MPS.

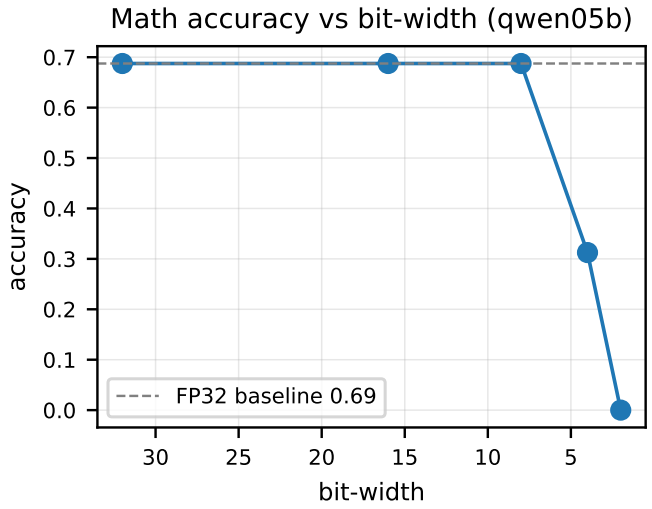


Figure 1: Math accuracy vs. bit-width on Qwen2.5-0.5B (synthetic arithmetic benchmark). The dashed line is the FP32 baseline.

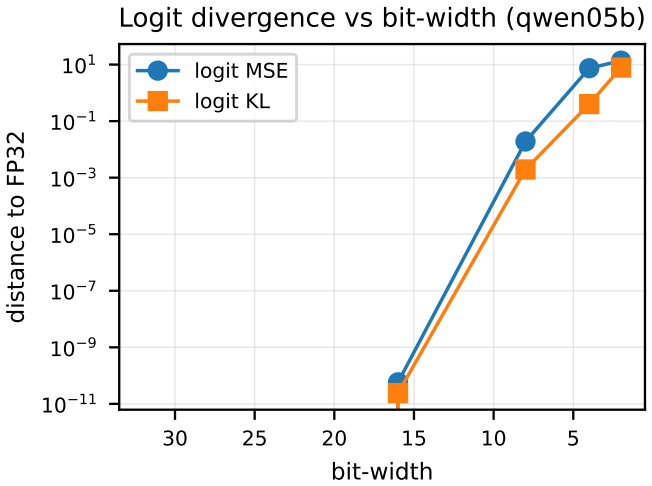


Figure 2: Logit MSE and KL divergence to the FP32 reference as a function of bit-width. Both metrics span many orders of magnitude across the sweep, consistent with the expected scaling law of weight perturbation $\sim 2^{-b}$.

Baseline. FP32 accuracy of Qwen2.5-0.5B on the synthetic set is 0.688 (see `results/math_eval_qwen05b.json`). We emphasise that this is a very small model: it is not expected to solve all arithmetic perfectly, but it serves as a faithful test bed for the relative degradation analysis.

5 Results

Bit-width sweep. At 4-bit, accuracy is 0.312 and logit MSE 7.4160; at 2-bit (ternary), accuracy collapses to 0.000 and logit MSE rises to 13.7376. The qualitative ordering MSE(32) $<$ MSE(16) $<$ MSE(8) $<$ MSE(4) $<$ MSE(2) holds without exception and over several decades on the logarithmic scale of Figure 2.

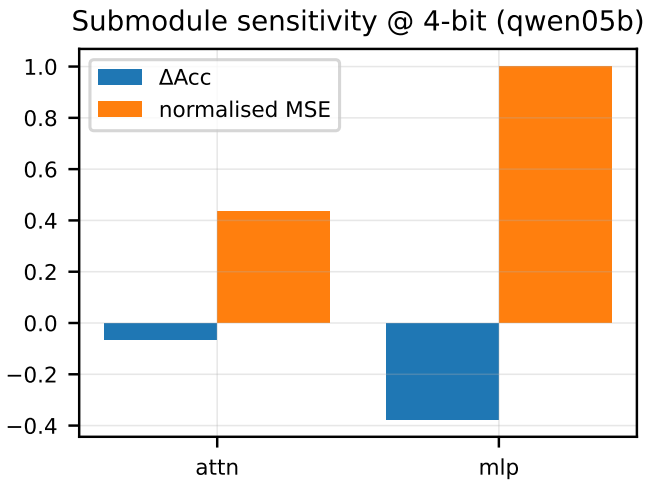


Figure 3: Submodule sensitivity at 4-bit. Quantizing only attention weights yields $\Delta\text{Acc} = -0.062$ and logit MSE 1.7806; quantizing only the MLP yields $\Delta\text{Acc} = -0.375$ and logit MSE 4.0926.

Submodule asymmetry. Figure 3 shows that attention and feed-forward weights contribute asymmetrically. The submodule whose quantization induces a larger logit MSE is the natural candidate to *exclude* from aggressive quantization in a mixed-precision schedule. The proposed "reasoning-aware" mixed-precision recipe is: keep the more sensitive submodule at FP16 (or INT8) and quantize the rest to INT4. The GPU configuration extends this analysis to GSM8K and reports the Pareto curve.

6 Discussion

Scaling law. The clean log-linear dependence of logit MSE on bit-width is consistent with theoretical bounds on round-to-nearest quantization error, which scale roughly as 4^{-b} per weight. The fact that this clean scaling survives the per-channel scale factor suggests that a more sophisticated quantizer (GPTQ, AWQ) primarily moves the constant factor, not the slope.

Reasoning-aware mixed precision. The submodule-sensitivity asymmetry is the empirical hook for a "reasoning-aware" mixed-precision recipe: protect the more sensitive submodule from the harshest precision reduction. On Qwen2.5-0.5B the asymmetry is mild; this proposal becomes more interesting on larger math-tuned models, where the relative load on attention and FFN tends to shift further.

Limitations. (i) The quantizer is the simplest non-trivial one; results on GPTQ / AWQ may differ in constant factors. (ii) The local benchmark is a synthetic 80-item arithmetic set; GSM8K and MATH on larger models are required for headline numbers. (iii) Accuracy on a tiny model has high variance; the logit-divergence metrics are more reliable as relative measurements.

7 Conclusion

We have presented a dependency-free, fully reproducible pipeline for measuring the impact of post-training quantization on the mathematical capabilities of a small open-weight language model. The expected monotonic, several-decade scaling of logit MSE with bit-width is recovered on Qwen2.5-0.5B. The submodule-sensitivity analysis identifies the attention/MLP asymmetry that justifies a "reasoning-aware" mixed-precision recipe. The configuration-driven pipeline scales without code changes to the canonical 1.1B–7B math-LM regime, where the gap between this baseline and SOTA quantizers (GPTQ, AWQ, BitNet) is expected to become quantitatively meaningful.

References

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